Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/30/2019
PI Name:	Rana, Brinda Ph.D.		
Project Title:	Proteomic Assessment of Fluid Shifts and Association with Visual Impairment and Intracranial Pressure in Twin Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	La Jolla	**State:**	CA
Zip Code:	92093-5004	**Congressional District:**	49
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	07/01/2014	**End Date:**	06/30/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/30/2018 per NSSC information (Ed., 6/22/17)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Ebert, Douglas Ph.D. (Wyle Laboratories, Inc.) Hargens, Alan Ph.D. (University of California, San Diego) Hook, Vivian Ph.D. (University of California, San Diego) Lee, Stuart M.S. (Wyle Laboratories, Inc.) Macias, Brandon Ph.D. (University of California, San Diego) Smith, Scott Ph.D. (NASA Johnson Space Center) Ziegler, Michael M.D. (University of California, San Diego) Patel, Hemal Ph.D. (University of California, San Diego) Schilling, Jan M.D. (University of California, San Diego) Stenger, Michael Ph.D. (NASA Johnson Space Center) Vaisar, Tomas Ph.D. (University of Washington) Hoofnagle, Andy M.D., Ph.D. (University of Washington)		
Grant/Contract No.:	NNX14AH58G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	This study is part of an integrated compilation of 10 studies, known as the NASA TWINS study, led by multiple Principal Investigators (PIs) which take advantage of a very unique opportunity to look at the effects of space travel on identical twin astronauts, one of whom experiences space travel for one year while the other remains Earth-bound for that same year. This investigation explores proteomic and genomic biomarkers underlying space flight-induced fluid shifts and visual impairment & intracranial pressure (VIIP) symptoms of astronauts in response to acute and prolonged exposure to microgravity and other space flight stressors. It examines pattern of changes in proteins before, during, and after the One Year Mission in the flight astronaut and compares these changes relative to the astronaut's twin brother on Earth. The overall goal of our study is to explore space flight related adaptations of proteins and genomic targets to further our understanding of the mechanisms underlying space flight-induced visual impairment/intracranial pressure (VIIP) symptoms and generate data on candidate biomarkers for the development of early detection methods for VIIP.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to the discovery of proteins involved in the evolution of space flight adaptations, this project has a broader impact on Earth-based clinical areas, such as traumatic brain injury-induced elevations of intracranial pressure, hydrocephalus, and glaucoma.
Task Progress:	Our study was part of an integrated, multi-center investigation that took advantage of the unique opportunity to look at the effects of a one-year space travel on an astronaut and compare physiological and molecular changes in his body relative to that of his identical twin brother, a retired astronaut who remained on Earth during the time course of the study. Specifically, for our study, we explored acute and long-term changes in proteins underlying exposure to a one-year space flight, focusing on symptoms related to spaceflight-induced fluid shifts and spaceflight-associated neuro-ocular syndrome (SANS). Using proteomics technology, we studied proteins in biofluids (blood and urine) before, during, and after exposure to space flight and compared protein profiles to that of the ground control twin astronaut. We also conducted physiological measures of the eye and the physiology of fluid compartmentalization in the body. We identified several changes in protein levels during space flight that correlated with physiological changes in the astronaut experiencing space flight. In particular, we observed a significant increase in the aquaporin 2 protein that corresponded with dehydration of the astronaut while he was in space. We also observed increased levels of collagen proteins during space flight relative to ground timepoints that we hypothesized was due to vascular remodeling that may occur as a result of long duration spaceflight. The physiology data also indicated that although some ocular (eye) changes recover in the post-flight, some persist after flight and may worsen with subsequent missions. The results of our study were published in Science in April 2019 as a "cover" article.
Bibliography Type:	Description: (Last Updated: 10/09/2024)
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." Science. 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650.long ; PubMed PMID: 30975860 , Apr-2019